

Feasibility Memo

The Yale Emergency Department triage dataset is suitable for developing a baseline machine learning model to predict Emergency Severity Index (ESI) levels. The dataset contains clinically relevant information collected during patient triage, including demographic characteristics, physiological vital signs and chief complaints, making it appropriate for exploratory modelling. However, because the dataset was collected from a single United States emergency department, any resulting model should be externally validated before being considered for deployment in Caribbean healthcare settings.

Dataset Summary

The dataset contains over two hundred clinical variables collected during emergency department triage, including approximately 200 binary chief complaint indicators together with structured demographic, administrative and physiological variables. The prediction target is the Emergency Severity Index (ESI), a five-level triage classification assigned immediately after patient assessment.

Exploratory data analysis showed that the dataset is well structured and suitable for analysis. Histograms of the physiological variables demonstrated realistic clinical distributions, while demographic plots showed the composition of the study population by race and ethnicity. Visualisation of ESI categories showed that most patients fall within ESI Levels 2 and 3, which is consistent with typical emergency department patient populations. Correlation analysis also identified meaningful relationships between several physiological variables and patient acuity, supporting their inclusion as predictive features.

Reasons to Proceed

1. Clinically meaningful predictor variables are available at triage.

The dataset contains variables routinely assessed by triage nurses, including age, heart rate, respiratory rate, blood pressure, oxygen saturation, temperature, blood glucose, arrival mode and chief complaints. These measurements are collected before treatment decisions are made, making them appropriate predictors for an AI-assisted triage model. Several vital signs also demonstrated observable relationships with ESI levels during exploratory analysis.

2. The dataset demonstrates good overall quality for baseline modelling.

Data exploration found that the structured variables were complete and internally consistent. Physiological measurements fell within expected clinical ranges, while exploratory visualisations confirmed reasonable distributions across demographic groups and triage categories. The chief complaint variables are encoded consistently

using binary indicators, providing a rich description of presenting conditions that can support predictive modelling.

3. The dataset contains diverse clinical information.

Unlike datasets that rely only on demographic or physiological variables, this dataset combines patient characteristics, vital signs and a large number of chief complaint indicators. This combination provides both objective physiological measurements and contextual clinical information that are expected to improve prediction of patient acuity compared with using vital signs alone.

Key Caveats

1. Limited external generalisability

The dataset originates from a single academic emergency department in the United States. Patient demographics, disease prevalence, healthcare resources and clinical workflows differ substantially from those in Caribbean hospitals. Consequently, model performance may decline when applied to local populations without additional validation and retraining.

Mitigation: Treat this work as a proof-of-concept and validate future models using Caribbean emergency department data before clinical deployment.

2. Outcome leakage must be avoided

Some variables, including patient disposition and previous disposition, represent outcomes that become known only after clinical assessment and treatment. Including these variables during model development would artificially inflate model performance while making the model unusable in real clinical practice.

Mitigation: Restrict model inputs to information available at the point of triage and exclude all post-triage outcome variables.

3. Class imbalance

The ESI distribution indicates that some triage categories occur much more frequently than others, particularly Levels 2 and 3. Without appropriate evaluation, models may achieve good overall accuracy while performing poorly on rare but clinically important high-acuity patients.

Mitigation: Evaluate performance using precision, recall and F1-score for each ESI category rather than relying solely on overall accuracy. If necessary, investigate class weighting or sampling techniques during model development.

Top-10 Feature Shortlist

Feature	Clinical Reasoning
Age	Older patients are generally at greater risk of severe illness and complications.
Respiratory Rate	Often one of the earliest physiological indicators of patient deterioration.
Heart Rate	Tachycardia may indicate shock, infection or cardiovascular compromise.
Oxygen Saturation	Low oxygen saturation reflects respiratory compromise requiring urgent assessment.
Systolic Blood Pressure	Hypotension is a key indicator of circulatory instability and shock.
Temperature	Fever or hypothermia may indicate serious infection or systemic illness.
Blood Glucose	Severe hypo- or hyperglycaemia can require immediate intervention.
Arrival Mode	Ambulance arrivals are frequently associated with higher-acuity presentations.
Chief Complaint Indicators	Presenting complaints provide important clinical context beyond physiological measurements.
Gender	Certain emergency presentations differ between males and females and may contribute modest predictive value while requiring fairness evaluation.

Conclusion

Overall, the exploratory analysis indicates that the Yale Emergency Department dataset provides a strong foundation for developing an initial machine learning model for Emergency Severity Index prediction. The combination of structured physiological measurements, demographic information and detailed chief complaint indicators makes the dataset clinically informative and appropriate for baseline modelling.

Although the dataset appears suitable for this research project, it should be viewed as a proof-of-concept rather than a deployment-ready clinical resource.

Github link : <https://github.com/Amanda-miller0/carisurg-portfolio.git>

